INSTITUTIONAL-GRADE PERSONAL FINANCE

# BUDGETMATE

A High-Fidelity **AI Ecosystem** for Institutional-Grade **Personal Wealth Management** — conversational logging, on-device vision, and live market telemetry in one premium Android experience.

| 5 | AI | 100% |
| --- | --- | --- |
| NATIVE MODULES | GEMINI + VISION | ON-DEVICE DATA |

BudgetMate · 2026

THE VISION

# Wealth tracking is fragmented. Cognition should be automated.

People juggle a banking app, a spreadsheet, a brokerage tab and a shoebox of receipts. BudgetMate collapses that into a single intelligent ledger that understands plain language.

### Today · The Friction

- ✕ Manual entry across 4+ disconnected tools, abandoned within weeks.
- ✕ Receipts lost; tax-deductible spend untracked.
- ✕ No link between daily spend and market exposure.

### BudgetMate · The System

- ✓ Type *"boba 6$"* — the LLM parses, categorises and books it.
- ✓ Vision OCR digitises any receipt in < 2 seconds.
- ✓ Spend, savings and equities on one telemetry surface.

FIRST IMPRESSION

# A splash worth screenshotting.

The launch sequence sets the tone: a metallic coin mark on deep-midnight glass, a four-card value tour, then straight into a frictionless first entry — no signup wall.

### Skeuomorphic gold branding

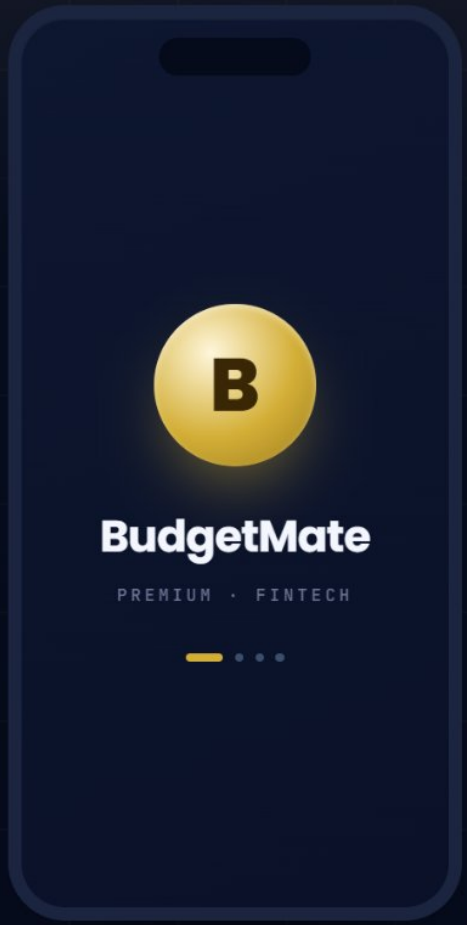
Embossed coin adaptive icon + animated splash with depth lighting.

### Four-slide value tour

Budget · Insights · News · AI — paged, dot-indexed, swipe-native.

### Zero-friction first log

Land directly on the keypad — value before account creation.

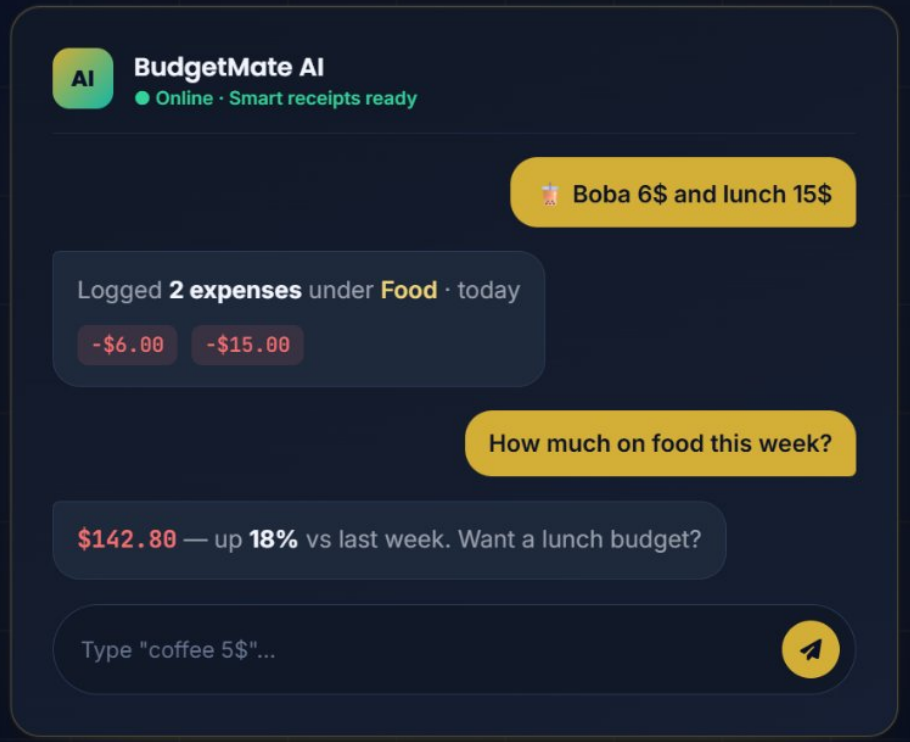


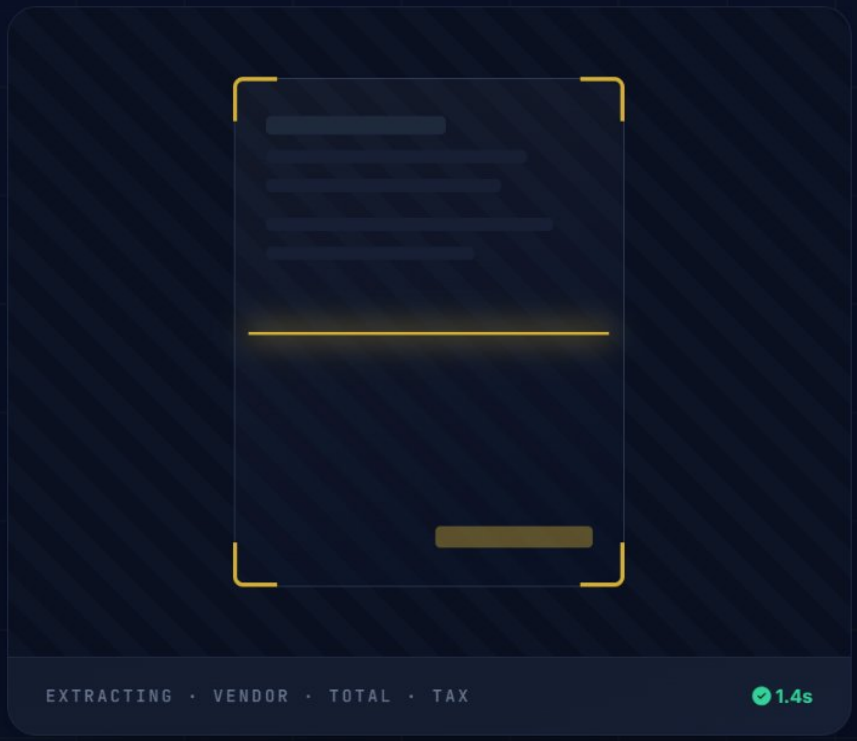
FEATURE · GOOGLE GEMINI

# Talk to your ledger.

A conversational layer powered by Gemini parses unstructured language into structured `ExpenseEntity` records — amount, vendor, category and note — then books them through the shared ViewModel.

- Natural-language parsing
- Auto-categorisation
- Quick-log chips
- Cashflow Q&A






FEATURE · ON-DEVICE OCR

# Snap. Parse. Booked.

A CameraX pipeline feeds an on-device vision model that lifts vendor, line totals and tax distribution straight off the receipt — no image ever leaves the handset.

### Vendor + total extraction

Structured fields mapped directly into the entry form.

### Tax distribution parsing

Separates net, tax and tip for accurate deductible tracking.

### Fully on-device

Processing stays local — zero cloud round-trip for imagery.

THE ANALYTICS CORE

# Income vs. Expense, visualised.




NET FLOW · MAY

**+$1,100**

AVG SAVINGS RATE

**24.6%**

CHART ENGINE

**MPAndroidChart**

MARKET TELEMETRY

# Live indices & equity nodes.

A decoupled ViewPager2 surface polls indices and Vietnamese tech equities — FPT, VIC, VCB — rendering anti-aliased cubic-Bézier trendlines tinted by daily change.

**FPT** +2.4%
FPT Corp
132,400



**VIC** -1.1%
Vingroup
45,150



**VCB** +0.8%
Vietcombank
91,800



**MWG** +3.1%
Mobile World
58,300



VN-INDEX · LIVE

1,287.4 ▲ 0.62%



Polling · 15s

Cubic-Bézier · anti-aliased

PRIVACY-BY-DESIGN

# Architectural sovereignty.

Financial data is a liability when it leaves the device. BudgetMate keeps the ledger sandboxed on-handset and treats credentials as build-time secrets — not runtime payloads.

### Room DB Sandbox

All transactions persist in a private SQLite store via Room — scoped to the app sandbox, never world-readable, never synced by default.

### BuildConfig Obfuscation

API keys live in `BuildConfig` + Gradle properties, stripped from source control and shielded by R8 obfuscation.

### On-Device Inference

Receipt vision runs locally; only opt-in conversational queries touch the network, and never raw imagery or the full ledger.

**Backup & extraction rules** are explicitly declared — the OS auto-backup surface is locked down so the ledger never lands in cloud backups unintentionally.

COMPETITIVE MATRIX

# Tracking app vs. cognitive mapping.

| Capability | Standard Tracker | BudgetMate Premium |
|---|---|---|
| Transaction entry | Manual forms | LLM natural-language parsing |
| Receipt capture | ✕ Photo storage only | On-device OCR extraction |
| Categorisation | Manual tagging | Automatic cognitive mapping |
| Market telemetry | ✕ Not available | Live indices + equity nodes |
| Financial insights | Static monthly totals | Conversational AI analysis |
| Data residency | Cloud-synced | On-device · Room sandbox |

ENGINEERING & CONTACT

# Built like a production system.

Native Android, MVVM with LiveData, and a decoupled data layer — engineered for stability, testability and premium feel.

### Java · Native Android

Fragment-based UI, programmatic ViewHolder generation, GridLayout keypad.

### MVVM + LiveData

Shared BudgetViewModel dispatches a reactive, lifecycle-aware data flow.

### Room · SQLite

Typed entities, DAO queries, LinkedHashMap chronological grouping.

### Retrofit · REST

Async market + news polling via decoupled ViewPager2 lifecycles.

### Gemini + Vision

Conversational parsing and on-device OCR receipt intelligence.

### MPAndroidChart

Cubic-Bézier sparklines, grouped bars, pie + radar analytics.